LED DOT-MATRIX SCROLLING DISPLAY Using STM32 Controller

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This project aims to display the information entered in a mobile phone on an LED dot-matrix type scrolling display. It is done by interfacing a Bluetooth HC-05 module to an LED dot-matrix display using an STM32 board.

The STM32 board, also known as Blue Pill development board, incorporates STM32F103C8T6 microcontroller from STMicroelectronics. It is a 32-bit ARM Cortex M3 controller with high clock frequency that is suitable for high-speed and power constraint applications.

LED dot-matrix is a widely used display that is seen in digital clocks at railway stations, airports, etc to show arrival and departure time of trains and airplanes. Components required for the project are listed under Bill of Material table. Author's prototype of the LED dot-matrix scrolling display using STM32 controller is shown in Fig. 1.

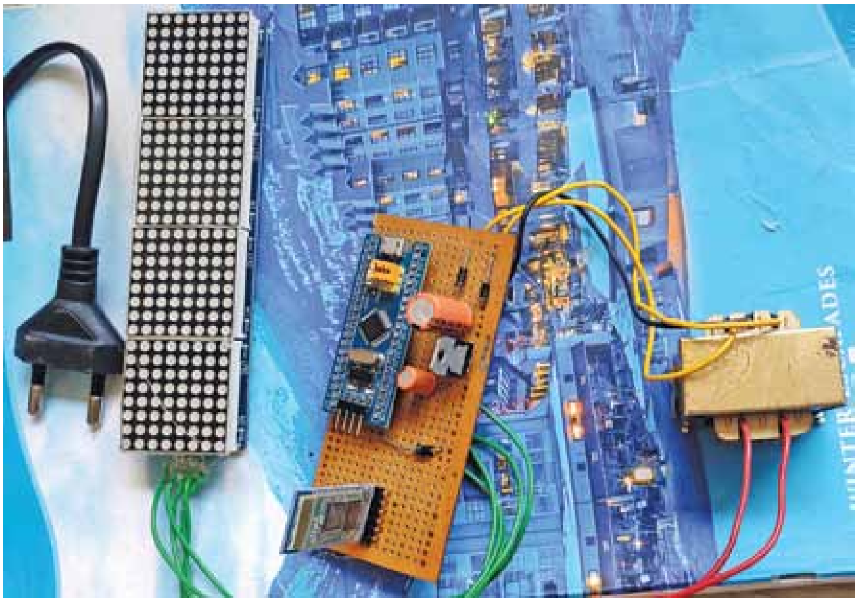
Fig. 1: Author's prototype of the display

As shown in Fig. 2, the dot-matrix LED module incorporates an MAX7219 driver chip. The module has an 8 × 8 LED matrix, each LED of which is controlled precisely by the MAX7219 driver to generate the colour pattern or text you wish to see. Thus, you have control over all the 64 LEDs by connecting the module just through three wires to microcontrollers like Arduino, NodeMCU, ESP32, STM32, etc.

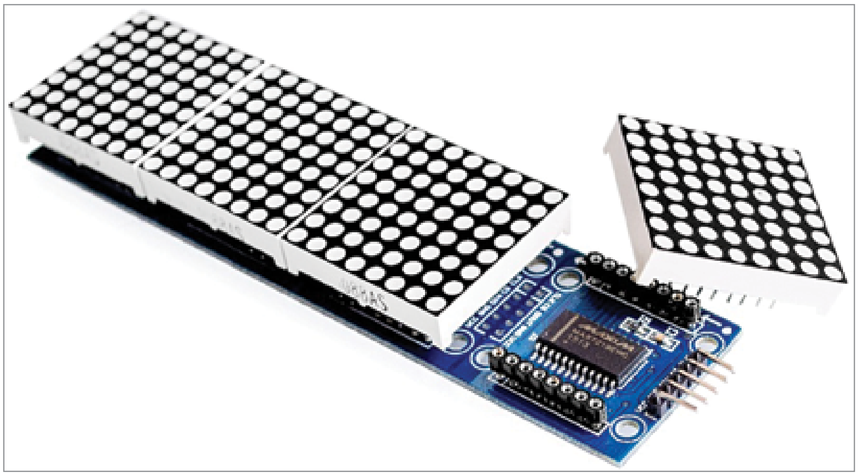
Fig. 2: MAX7219 LED dot-matrix 4-in-1 display module

This module has four dot-matrix LEDs cascaded to form a single LED display to allot more area to the user on the display. Connecting two such modules is easy; just connect output pins of the previous display to input pins of the next display. This way any number of dot-matrix LED modules can be connected to an Arduino, NodeMCU, ESP32, or STM32.

TABLE 1
BILL OF MATERIAL

- MAX7219 LED Dot Matrix Display (1088AS)
- STM32 (Blue Pill) Development Board
- HC-05 Bluetooth Module
- 5V Regulated Power Supply
- General-Purpose PCB/Bread board
- Jumper Wires